

Simulation: DMSO distillation

BATCH DISTILLATION 1 CALCULATION RESULTS

Operation Step 1:

STREAM PROPERTIES

Name	Pot Charge	Accumulator	Pot Residue	Distillate
- - Overall - -				
Molar flow kmol	109.6953	14.7901	93.8819	0.1129
Mass flow kg	7669.8066	268.6261	7329.7456	3.5212
Temp C	25.0000	45.8518	118.5419	53.4486
Pres bar	0.1000	0.1000	0.1000	0.1000
Vapor mole fraction	0.000	7.811E-05	0.000	0.000
Enth MJ	-29012.	-4201.2	-23145.	-31.341
Tc C	449.5155	374.8857	452.8278	413.6053
Pc bar	84.6829	222.0417	56.6975	230.5752
Std. sp gr. wtr = 1	1.100	1.001	1.104	1.055
Std. sp gr. air = 1	2.414	0.627	2.696	1.077
Degree API	-2.8940	9.8589	-3.3614	2.6606
Average mol wt	69.9192	18.1625	78.0741	31.1975
Actual dens kg/m3	1091.6582	337.3329	1002.7520	1029.6150
Actual vol m3	7.0258	0.7963	7.3096	0.0034
Std liq m3	6.9709	0.2684	6.6376	0.0033
Std vap 0 C m3	2458.6716	331.5005	2104.2358	2.5298
- - Vapor only - -				
Molar flow kmol		0.0020		
Mass flow kg		0.0357		
Average mol wt		18.0163		
Actual dens kg/m3		0.0680		
Actual vol m3		0.5252		
Std liq m3		0.0000		
Std vap 0 C m3		0.0444		
Cp kJ/kg-K		1.8693		
Z factor		0.9987		
Visc N-s/m2		1.038e-05		
Th cond W/m-K		0.0199		
- - Liquid only - -				
Molar flow kmol	109.6953	14.7881	93.8819	0.1129
Mass flow kg	7669.8066	268.5904	7329.7461	3.5212
Average mol wt	69.9192	18.1626	78.0741	31.1975
Actual dens kg/m3	1091.6582	990.6572	1002.7521	1029.6150
Actual vol m3	7.0258	0.2711	7.3096	0.0034
Std liq m3	6.9709	0.2683	6.6376	0.0033
Std vap 0 C m3	2458.6716	331.4561	2104.2358	2.5298
Cp kJ/kg-K	2.0360	4.1532	2.0817	2.9660
Z factor	0.0004	0.0001	0.0003	0.0002
Visc N-s/m2	0.001861	0.0005881	0.0005887	0.0006244
Th cond W/m-K	0.1644	0.6281	0.1242	0.3031
Surf. tens. N/m	0.0447	0.0681	0.0308	0.0513
Component mole %				
Dimethyl Sulfoxi	86.334318	0.245411	99.898696	21.927060
Water	13.665682	99.754590	0.101302	78.072941

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